UC DAVIS VETERINARY MEDICINE

Clinical Anatomy

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We won't cover every aspects of the anatomy of small mammals as it is very similar to that of other domestic mammals. We will mainly **focus on clinical anatomy meaning we will highlight anatomical and physiological features of high clinical relevance**. We won't talk a lot about the ferret because its anatomy is mostly the same as dogs and cats. Regarding hedgehogs and sugar gliders, there is a dedicated lecture on these two species, which will also cover their clinical anatomy. So we will concentrate on rabbits and rodents in this lecture. Finally, as there is an entire lecture on rabbit and rodent dentistry, we also won't talk about their dental anatomy, except when it affects other organ systems.

There is also a **lab**, which will highlight some other important aspects of small mammal anatomy.

The learning objectives are:

-Compare and contrast clinically relevant anatomy amongst small mammals.

-Recall important anatomic differences that can affect disease presentation and veterinary procedures.

SKIN

Rabbits

Rabbits' skin is **thin and fragile** and easily traumatized. It may be an adaptation to escape predators. Clinically, it is particularly an issue when using clippers and **clipper burns and cuts** can easily be induced.

Mature unspayed does (female rabbits) have a **dewlap**, which is a flap of skin under the chin. It can be pretty big in some breeds. Rabbits **pull hair from their dewlap** to use as a nesting material just prior to parturition. In some circumstances (obese rabbits, ptyalism, poor hygiene), a moist dermatitis may develop at that location.

Rabbits have dense fur on their plantar surface and metatarsus that is used as cushioning in place of digital/metatarsal pads like in carnivores. **DO NOT shave the hair on the plantar surface of the hind feet** or rabbits will develop pododermatitis.

Rabbits do not have sweat glands and thus dissipate heat through other mechanisms (ear, panting). This, among other things, make rabbits **highly susceptible to heat stroke** in poorly ventilated and hot areas (especially when temperature is higher than 85F). However, they have a number of scent glands including under the chin (you can frequently see them rubbing their chin on things), anal, and inguinal scent glands. The most important to remember are the **inguinal scent glands** as they can become impacted and kind of gross.

Rodents (and sugar gliders)

The main unique feature of this groups is the presence of scent glands and their location, which vary widely depending on the species.

Syrian hamsters have one on each flank, while gerbils (and dwarf hamsters) have one midline ventrally (common site of neoplasia in gerbils), guinea pigs have one on the rump (also called grease gland) and sugar gliders have one on the top of the head. Those should not be confused with a skin lesion, that's why it is important to know their location by species.

Also another interesting fact in **gerbils: the skin of their tail can slough off** when grabbing the tail (escape mechanism), so don't grab their tail.

Ferrets

The main thing to know about ferrets when it comes to skin physiology is that there is a fairly marked seasonal effect. **Ferrets molt heavily in the fall and spring**. The coat is shorter in summer and longer in the fall. The molt can be dramatic in the spring and lead to **seasonal alopecia of the tail**.

During the breeding season, intact ferrets have increased sebaceous secretions that leads to more of a yellowish coloration. Shedding is associated with a concomitant change in body weight (heavier in winter). **Adrenal gland disease (neoplasia) is very common in spayed/neutered ferrets and leads to bilateral symmetric alopecia** and can be associated with a prolonged alopecia of the tail as well.

Most ferrets are descented and neutered and they are tatooed on the ear as a result (one dot for each procedure performed).

Ferrets have very active sebaceous glands that are mainly responsible for their **musky odor**, more so than their anal glands. Neutering decreases overall musky odor, but does not eliminate it.

Fur Coat Colors, Types and Health

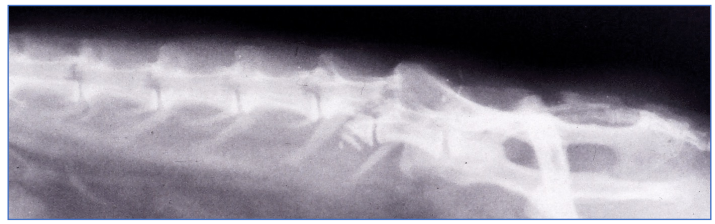
There are a number of fur colorations and types that are associated with medical conditions in small mammals. For instance, white rabbits with blue eyes have an increase prevalence of idiopathic epilepsy.

The satin guinea pigs are predisposed to fibrous osteodystrophy (see musculoskeletal lecture) and skinny pigs, lacking hair, are prone to hypothermia.

Breeds are not officially recognized in ferrets, but deafness is associated with certain color patterns, most notably the panda and blaze types.

MUSCULOSKELETAL SYSTEM

Rabbits have an extremely light skeleton that predisposes them to orthopedic issues. It is estimated that only 8% of their total body weight is made of bones, compared to 12-13% for instance in cats. They also have a **well developped epaxial musculature and hind legs** for jumping. As a consequence, **rabbits can break their back** when trying to jump or when they are inappropriately restrained. The fracture most commonly happen at the **lumbo-sacral junction** (see neurology lecture).

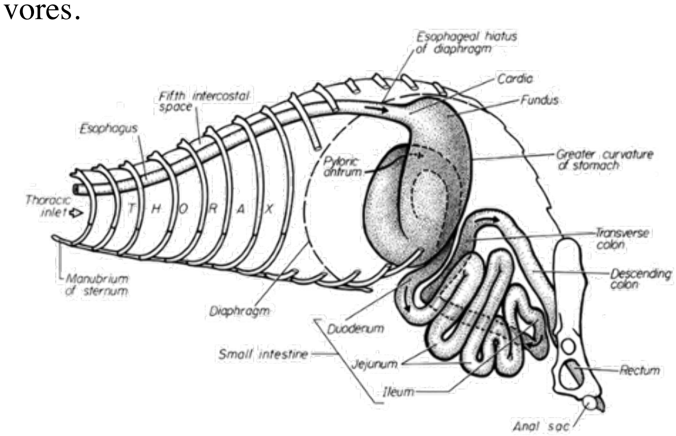


The long bones of the pelvic limbs are also prone to fractures, even more so in chinchillas (see musculoskeletal lecture).

GASTROINTESTINAL SYSTEM

Ferrets

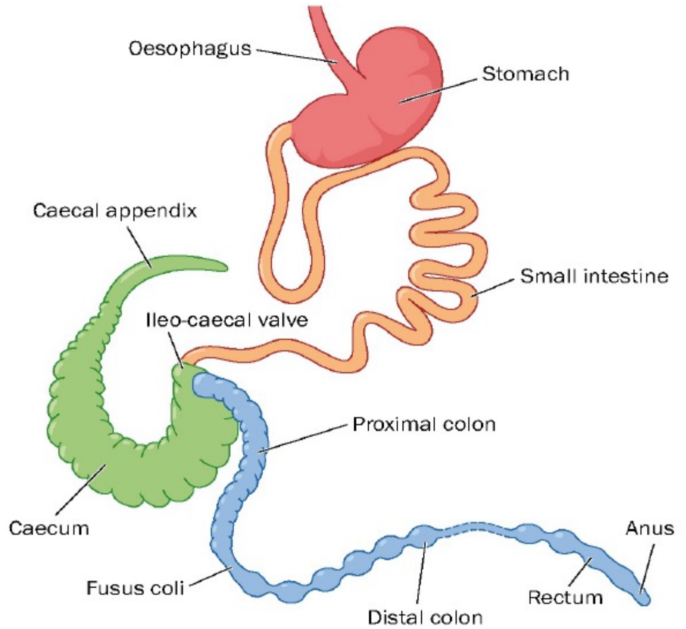
We won't spend too much time on the gastrointestinal system of ferrets as it is fairly similar to other carnivores.



Ferrets have an extremely **short transit time at 3-4h**, so they don't need to be fasted for long. The **stomach can distend a lot** accomodating the preys they eat. The jujunum and ileum are indistinguishable macroscopically and there is no cecum in the large intestines. Ferrets tend to have **prominent abdominal lymph nodes**, which is important to know when interpreting the results of an abdominal ultrasound examination. The **gastric lymph node** in particular often appears enlarged, which may or may not be abnormal. **Mesenteric lymph** nodes may also appear bigger than expected. These abdominal lymph nodes (especially mesenteric) are common sites for **lymphoma, a common tumor of ferrets**.

Rabbits

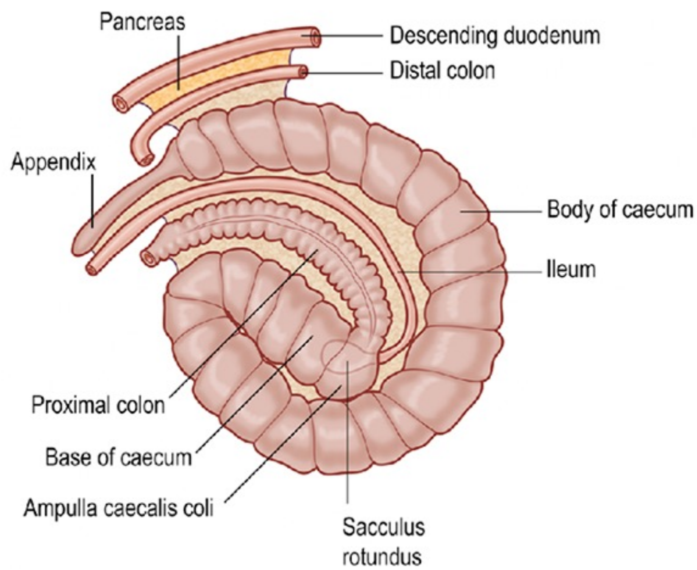
Knowing the anatomy and physiology of the gastrointestinal system of rabbits is very important as they are **prone to gastrointestinal diseases such as GI ileus, GI obstruction, GI dysbiosis, and cecal impaction**.



Rabbits are monogastric herbivores and the **stomach mainly acts as a reservoir of food. The stomach never fully empties** and it would take 9 days of continuous fasting to have it completely empty. Rabbits are **unable to vomit**. They have a powerful sphincter at the cardia, but also lacks the neurologic reflexes necessary for emesis. Consequently, there is no need to fast rabbits prior to anesthesia. One hour of fasting is still recommended to clear the oral cavity. The pH of the stomach is very acid (pH=2) with HCl mainly produced by the stomach fundus. The **pylorus is well developped and the dudenum exits the stomach at a very acute angle**. That predisposes the duodenum to be compressed with stomach distention and with trichobezoar, which can lead to mechanical obstruction.

The small intestinal anatomy is fairly standard. Small intestines are short in rabbits and are where the pancreatic and bile duts empty. Peyer's patches (lymphoid tissues) are present in the ileum.

The large intestines start at the **cecum**, which is huge in rabbits and where hindgut fermentation occurs. The ceum contains about 40% of intestinal content, so about 10 times more than the stomach. The ileum empties in the cecum at the ileo-cecal junction, which also sees several lymphoid tissues, namely the **sacculus rotundus** and the ileocecal plaque. It also terminates into a **vermiform appendix**, which is also a lymphoid organ and can get severily inflammed in some cases (called **appendicitis**). The sacculus rotundus may be misinterpreted for a mass on ultrasound. These lymphoid organs may also be sites for lymphoma. The body of the cecum itself has thin walls and presents a high number of bulges in a spiraling pattern (refers to as the spiral fold or **spiral valve**). The cecum folds 3 times on itself when in normal topographical location within the abdomen.

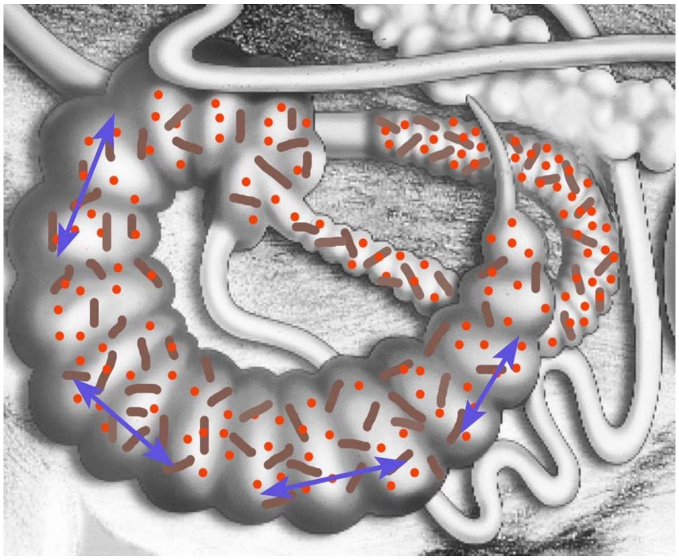


The colon is interesting in rabbits. It is structurally and functionally divided into 2 sections. The first section is the **proximal colon which is extensively sacculated into "haustras"** and emerges from the ampulla coli of the ileo-cecal area. The **fusus coli** is the thickened portion which separates it from the distal colon. It is heavily supplied by ganglion cell aggregates and actually is the **pacemaker of the hingut fermentation cycle**. The second colonic section is the distal colon which is less sacculated and where feces are located.

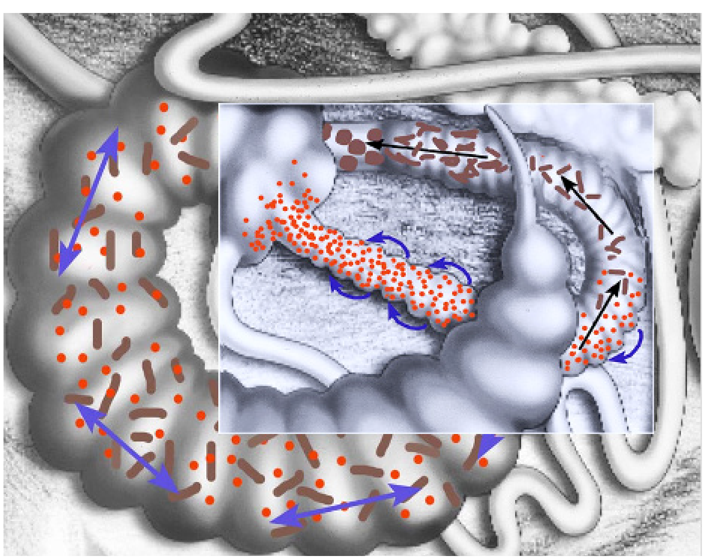
6: poximal colon, 7: fusus coli, 8: distal colon

So the way that whole thing works is as follow (also see this video):

Food composed of digestible and non-digestible fibers arrive to the cecum by the ileum in which it is constantly mixed and fermented by the cecal microflora. This produces **volatile fatty acids** that are absorbed through the cecal wall. **These volatile fatty acids constitute about 30-50% of the rabbit basal metabolism**.



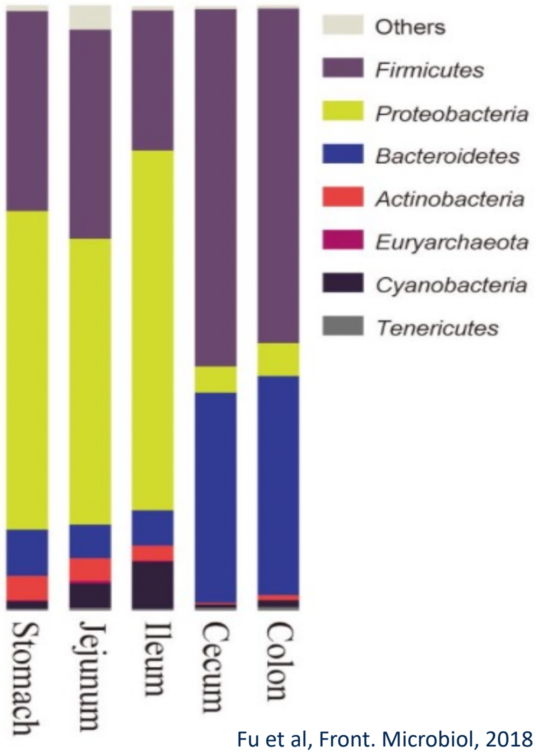
Then cecal content goes into the **proximal colon where there is a selective retention system** during which fine particles are entraped inside mucosal protrusions (called warsen or warts) and brought back along with fluids to the cecum by retroperistalsis. **Larger particles of non-digestible fibers keep going to the fusus coli. In the fusus coli, particles are compacted into pellets rich in these non-digestible fibers**. Even though non-digestible fibers do no contribute metabolizable energy, their bulk properties promote gastrointestinal motility, cellular regeneration, and muscle tone and help regulate microbiota composition. So they are very important in the rabbbit's diet.



But wait, there is another twist to this GI physiology story. **Rabbits produce two types of poops: hard and soft feces and they eagerly eat their soft feces, also called cecotrophs**. Cecotrophy is different from coprophagy, because they only eat a certain type of feces. This is their strategy to benefit from the product of fermentation though another mechanism than volatile fatty acid absorptions. **The cecotrophs are high in proteins and vitamins (mainly vitamin Bs and K).** They are coated by a mucus layer in the colon that protects them from stomach pH and they are digested mainly in the small intestines. They are also known as "night feces" (some people prefer the term of midnight snacks) as they are not produced right after eating but at specific intervals. They look different as they are blacker and more moistened and stuck to the perianal area while regular fecal pellets are hard and brown.

They are consumed directly from the anus. Consequently, placing an E-collar on a rabbit will prevent cecotrophy and sick rabbits may not consume them (or if their back hurts for instance).

The **microflora of the GI** is very diverse and unique to rabbits. Most of the bacteria belong to the firmicutes (clostridium-like bacteria) and bacteroidetes phyla. There are lots of Archae as well as some yeasts such as *Cyniclomyces guttulatus* (cecum and stomach), *Candida* spp. and *Malassezia* spp. Some ciliate protozoans are also present. There are no *Lactobacillus* (so you can ditch yogurt as a probiotic for rabbits). **The microflora changes drastically from the stomach/small intestines to the cecum/colon. Dysbiosis is common in rabbits**, which is characterized by a a disruption in this fine balance or the emergence of opportunistic pathogens such as *Clostridium* spp. It is promoted by the use of selected antibiotics that kill the normal gut flora but also by the lack of non-digestible fibers in the diet and some other gastrointestinal diseases.

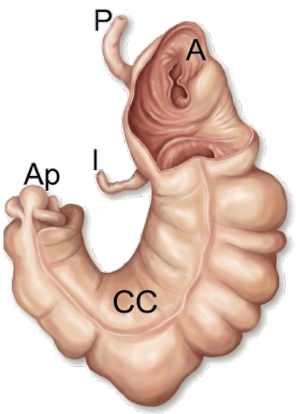


Fu et al, Front. Microbiol, 2018

Rodents

Let's start the rodent section by introducing the **cheek pouches of hamsters**. They are just incredibly huge. They can literally stuff them with about 20% of their body weight as food and they extend as far as the mid-body or more. There are muscles associated with them. Clinically, we see several issues with cheek pouches such as impaction, prolapse, and cheek pouch neoplasia. **Always inspect them** during physical examination if you can (for instance under anesthesia).

Alright let's move on to the rest of the GI. Like rabbits, **rodents are monogastric and hindgut fermenters**. However, their overall gastrointestinal strategy to process grass and plants is less effective than rabbits. They do not have a selective retention system in the colon for instance. They only have a **"mucus trap"** to send back bacteria and small soluble particles to the cecum by retroperistalsis. As a consequence, **herbivorous rodents have a much larger GI than rabbits** to make up for this. The cecum of guinea pigs contains 65% of the intestinal content and the colon is also larger than in rabbits. The cecum does not have a sacculus rotundus or vermiform appendix. The cecum is so big that it is likely the **first organ you will see in guinea pigs upon entering the abdominal cavity**.

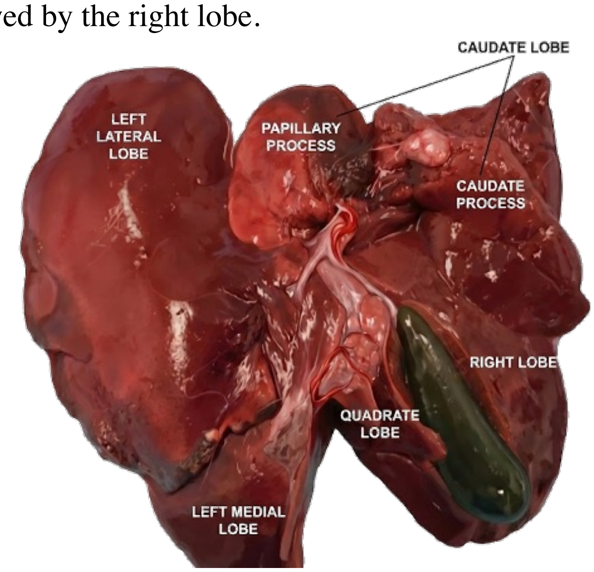


Cecotrophy and coprophagy have not been well investigated in hystricomorph rodents. Prevention of coprophagy leads to weight loss, so it is still an important process in them. See also this video.

Unlike guinea pigs and chinchillas, **rats and mice (myomorph rodents) are omnivorous**. In myomorph rodents, the stomach has a **limiting ridge** separating the forestomach from the glandular stomach (see Anatomy lab). This also prevents rats from vomiting.

LIVER

Rabbits are **prone to liver lobe torsion**, so you need to know lagomorph liver lobe anatomy to be able to identify which liver lobe is torsed. The rabbit has 4 main liver lobes: the caudate lobe (divided into papillary process and caudate process), the right liver lobe (where the gallbladder is), left liver lobe (divided into medial left and lateral left) and the quadrate liver lobe. **The caudate lobe is the most commonly torsed** followed by the right lobe.

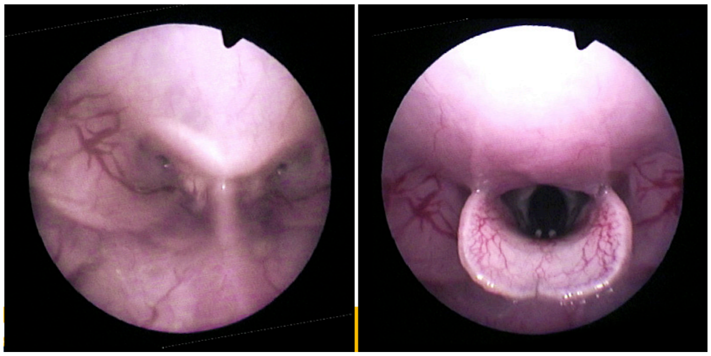


Note that the rat does not have a gallbladder contrary to most often small mammals species seen as pets including ferrets, rabbits, and most pet rodents.

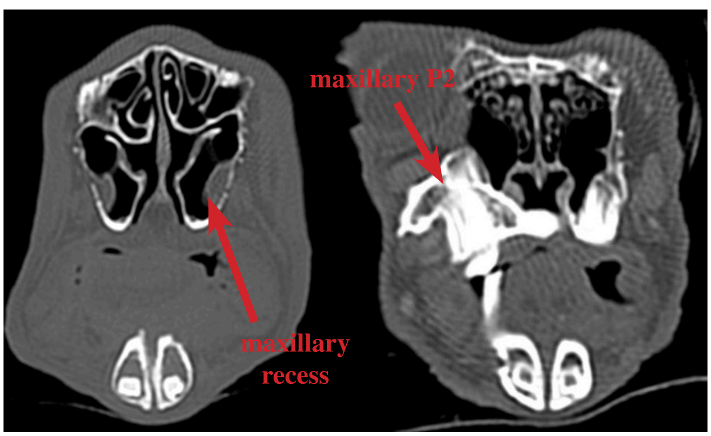
RESPIRATORY SYSTEM

In this section, we'll only talk about rabbits and rodents. The ferret is pretty similar to dogs and cats so there's nothing special to highlight for this species.

Rabbits are obligate nasal breathers. The epiglottis is dorsal to the soft palate at rest.



The glottis is also further caudal and hard to see. This means that **any disease of the upper respiratory system will result in dyspnea**. **Odontogenic rhinitis/sinusitis in rabbits is common** because the first maxillary premolar (P2) is very close to the **ventral recess of the maxillary sinus**. Retrograde elongation of its reserve crown tends to perforate the maxillary recess of the maxillary sinus, which is where most odontogenic rhinitis start. See the CT cross-sectional images below (left is more rostral than right).

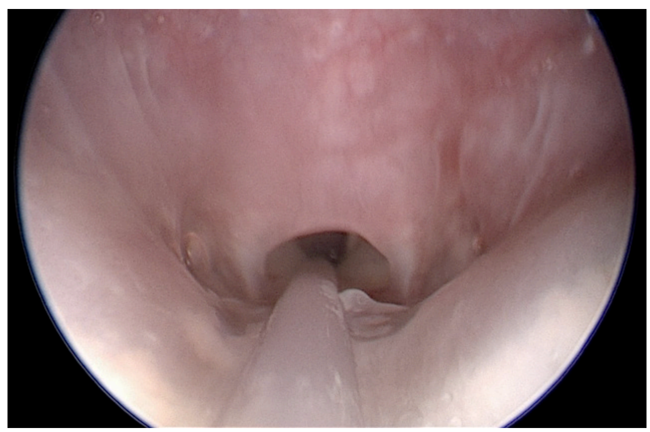


The second clinical application is that **rabbits are hard to intubate**. We can intubate them blindly or using visual aids such as an endoscope.

Rabbits have **3 paranasal sinuses: the dorsal conchal sinus, the maxillary sinus (divided into ventral and dorsal recesses), and the sphenoidal sinus** (see Anatomy lab handout).

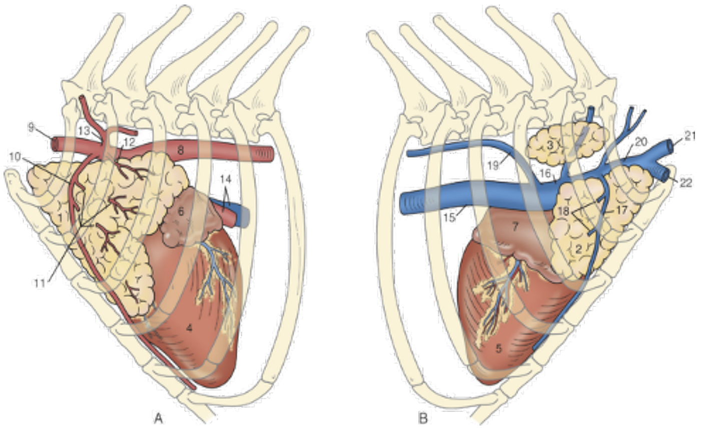
Rodents are also obligate nasal breathers. Like rabbits, any disease or mass in the upper airways will lead to dyspnea. An example of such a mass is **elondotomas** (pseudotumor of odontogenic tissue), which are common at the apex of maxillary incisors in prairie dogs and degus. Another anatomical feature of hystricomorph rodents is the presence of a **palatal ostium**, which results from the **fusion of the soft palate to the base of the tongue.**

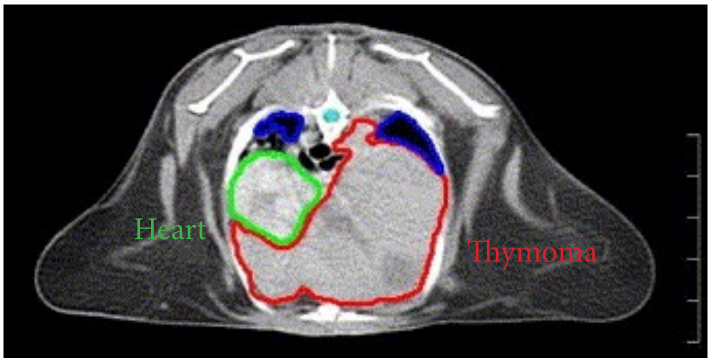
This means that rodents are even harder to intubate than rabbits. We can use an endoscope to place a stylet into the trachea prior to threading an ET tube.



Rabbits and rodents also have a small chest and thoracic cavity, especially when comparing to their ginormous abdominal cavity. **The lungs are divided into lobes, but not into lobules** like in dogs. This means that pneumonia will result in lobe consolidation and not a lobular pneumonia.

Last thing of importance is in rabbits. **Their thymus persists into adulthood and is located in the cranial mediastinum**. Why do we care? We in fact care deeply because **thymomas and thymic lymphomas** are exceedingly common in rabbits and will result in a large cranial mediastinal mass that will push the heart and the trachea (see CT radiation planning image below).

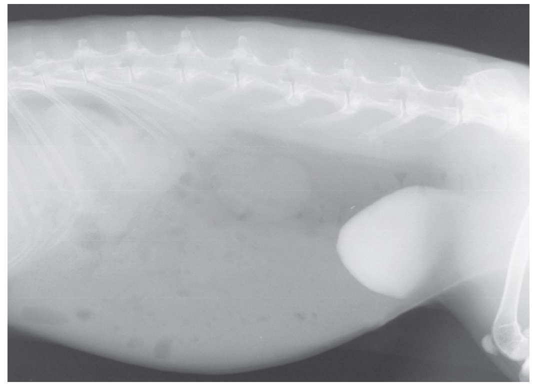




URINARY SYSTEM

The urinary system is similar to other mammals. One thing worth knowing is about the calcium metabolism of rabbits, which is very unique.

Rabbits absorb most dietary source of calcium from their gastrointestinal track. **Intestinal calcium absorption is not regulated by vitamin D like in other mammals so plasma calcium is in fact a direct reflection of dietary calcium intake.** They supposedly adapted to do that to support dental continuous growth. In addition, unlike most other mammals which excrete calcium in the urine at a 2% fractional excretion, **rabbits have as much as 45-50% fractional excretion of calcium in their urine**. When a lot of calcium is in the urine, which is physiological, this can appear as "**sludge**" on xray. Urine in the cage will look milky and thick. As it is related to calcium consumption, reducing calcium intake is recommended.



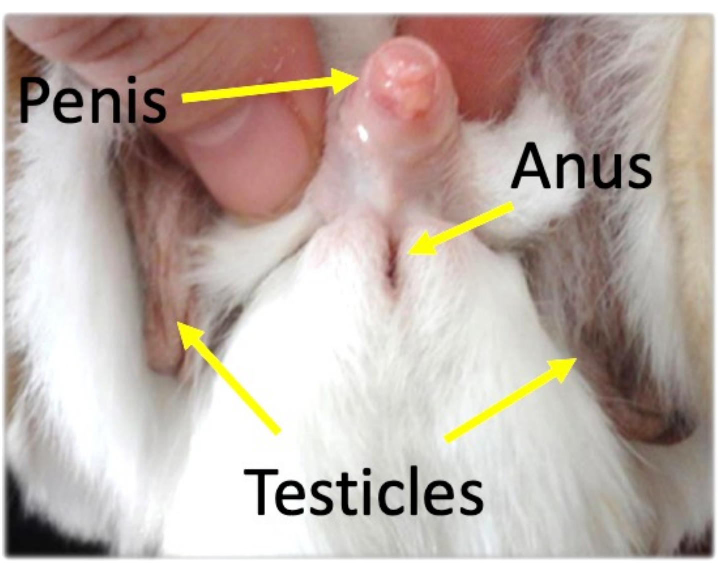
Since rabbits and rodents are herbivores, their urinary **pH is high, around 8-9**.

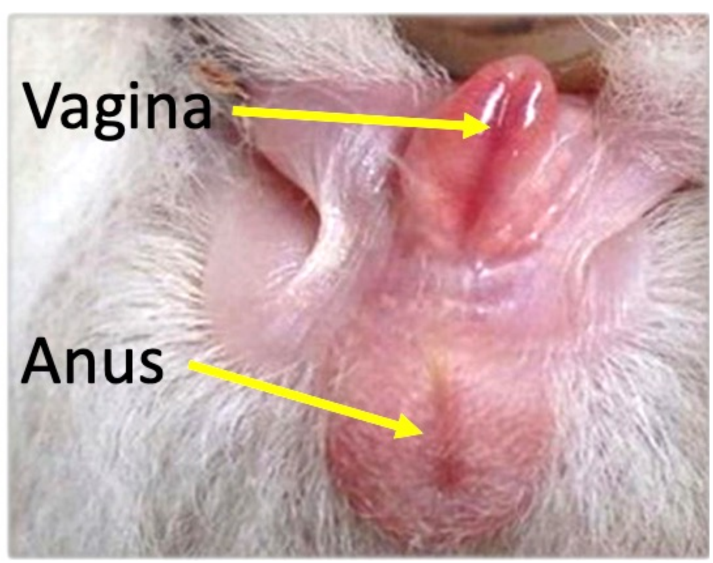
Last thing about the urine is the fact that r**abbits and rodents may excrete porphyrins, which will result in orange-to-red urine**. It should not be confused with hematuria. The color is more dark orange than bright red like blood, so most of the time you can readily tell the difference. We don't always know where these porphyrins come from, but supposedly from the breakdown of bile, plant pigments, some veggies, and some drugs.

REPRODUCTIVE SYSTEM

Sexing

Sexing rabbits and rodents may not be hard when they are adult and testicles are visible and palpable, but it is much harder when they are juveniles. That's typically when you would be asked to sex them because owners

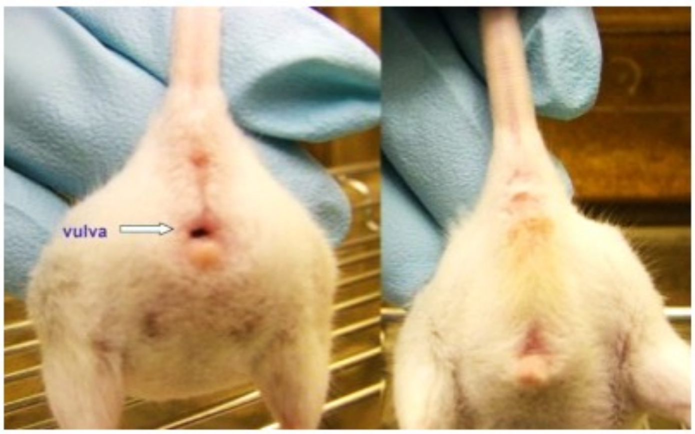




don't know how or would assume the wrong sex.

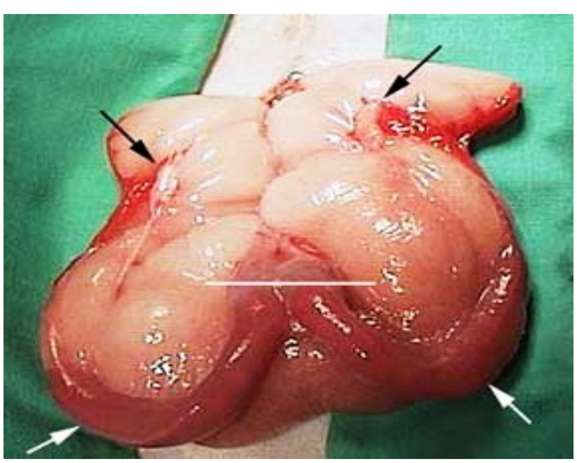
In rabbits (see pictures on previous page), when testicles have not yet descended (at about 3 months), you need to evert the genitalia. **The female has a prominent vulva with a slit-like opening while the male has a small penis with a roundish opening**. Sex your rabbits during the labs and you'll see, it's not as obvious as it may seem.

Guinea pigs and chinchillas are easy to sex by everting their penis (or looking for their scrotal sac/scrotum and testis if old enough). In other rodents such as rats and mice, the best way to sex them when they are immature is to compare the **ano-genital distance between sexes**. It is much longer in males than in females.



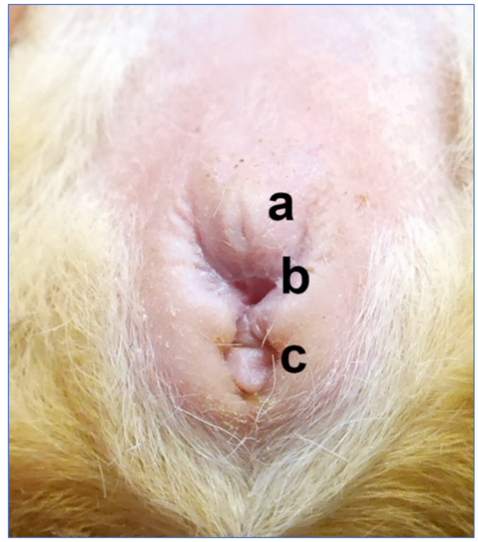
Female Anatomy

Rabbits have a bicornuate duplex uterus, meaning **two uterine horns with two cervices**. They also have a long and flaccid vagina. We in fact remove part of the distal vagina during spaying and perform an **ovario-hysterovaginectomy**. **The mesometrium is a major fat storage organ in the rabbit**. It can be an issue during spaying in older does as it makes it harder to identify the blood vessels during surgery and makes the mesometrium more friable as well.



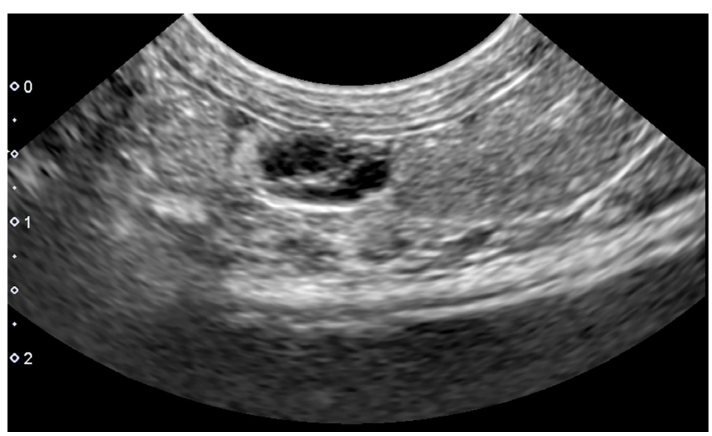
Unspayed rabbits have a very **high prevalence of uterine adenocarcinoma** (like 60-70% of them).

In guinea pigs and chinchillas, the uterus has two uterine horns with one (guinea pigs) or two cervices (chinchillas). **In rodents, the urethral opening is located outside of the vagina** (a on the figure below). It is therefore fairly easy to catheterize.

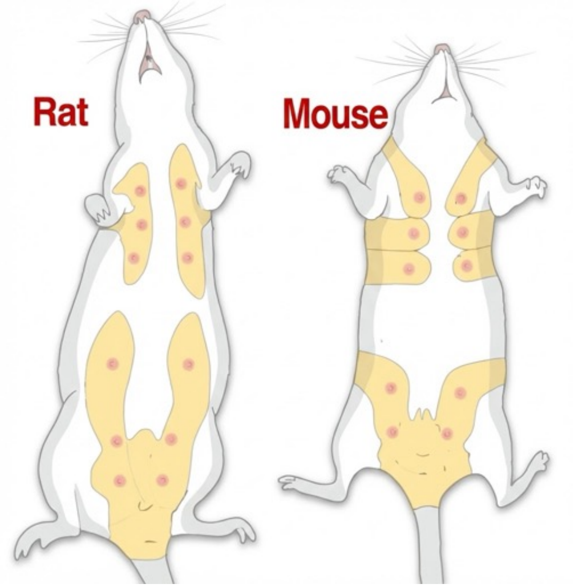


There is also a **vaginal closure membrane in rodents** that seals the vagina at all time except during estrus and parturition. Also a **vaginal plug** is present after copulation that is made from male secretion. It tends to persist for less than a day.

The female guinea pig (sow) has also a number of reproductive conditions that are worth mentioning here. **If the sows are not bred before 7-9 months of age, there will be a permanent incomplete separation of the pubic symphysis**, which may lead to dystocia during parturition after that age. Sows are also **prone to ovarian cysts** (see ultrasound figure of a small cyst here).



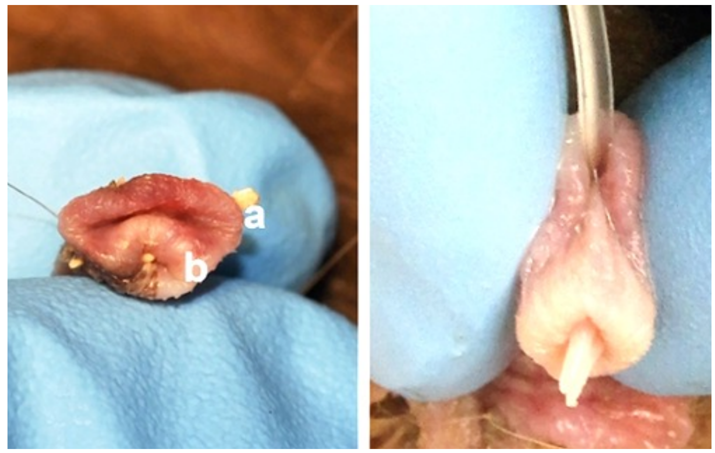
In rats and mice, the mammary tissue is very extensive and even goes dorsally towards the shoulders and hips. It is important to know because mammary gland tumors are common and it means that mammary tumors may occur almost anywhere on the trunk and proximal limbs.



We have not talked too much about ferrets because they are so similar to other domestic carnivores, but there are a few important things to know. As was mentioned multiple times, most ferrets are spayed prior to purchase so some of these issues will only be seen on ferrets bought through small private breeders (or in European ferrets, which are not spayed as breeding is less centralized). **Ferrets are induced ovulators**. Female ferrets (called jills) will not ovulate spontaneously and continue to produce estrogen, which can have some severe consequences. Unspayed jills can then develop **hyperestrogenism, which in turn will lead to aplastic anemia**. This can also happen with **ovarian remnants**, which are diagnosed from time to time. Jills in estrus will also have a considerably **swollen vulva. This swollen vulva can also be seen with adrenal gland disease (see below).**

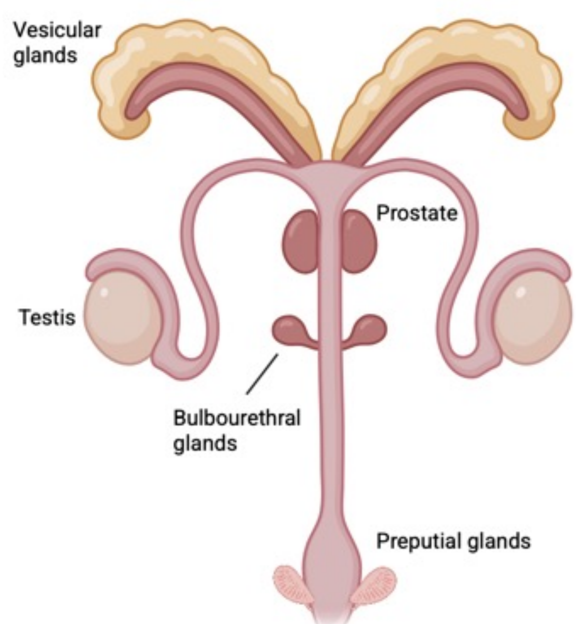
Male Anatomy

In rabbits and rodents, the penis is located caudal to the testis. Rabbits do not have an os penis but rodents do. Something very peculiar is seen on the rodent's penis. **Guinea pigs and chinchillas have a small intromittent sac at the penis tip that is lined with spurs when erected.** This should not be confused with the urethral opening further proximal for catheterization.

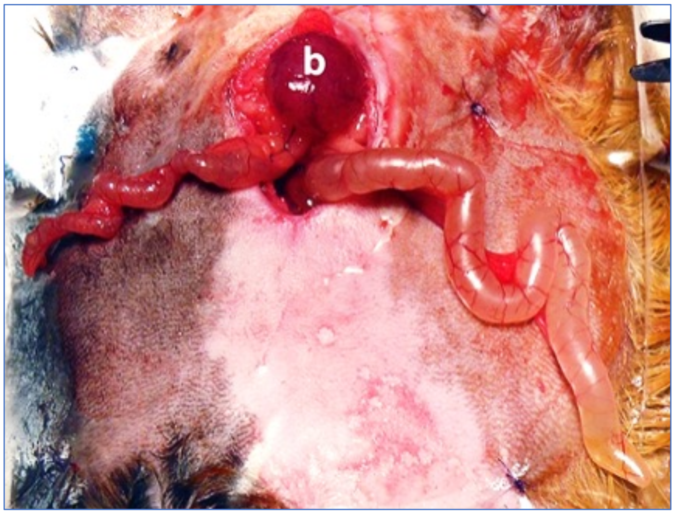


The testes descent at about 3 months and the inguinal canals remain open in rabbits and rodents. That means that they can pull their testicles back into their abdomen. Clinically, this also allows castration through a laparotomy approach (recommended in guinea pigs, see surgery lecture). Furthermore, scrotal hernia is possible through that open iguinal canal. In general the scrotum is not well developped in some rodents (guinea pigs) and is more of a scrotal sac than a true scrotum.

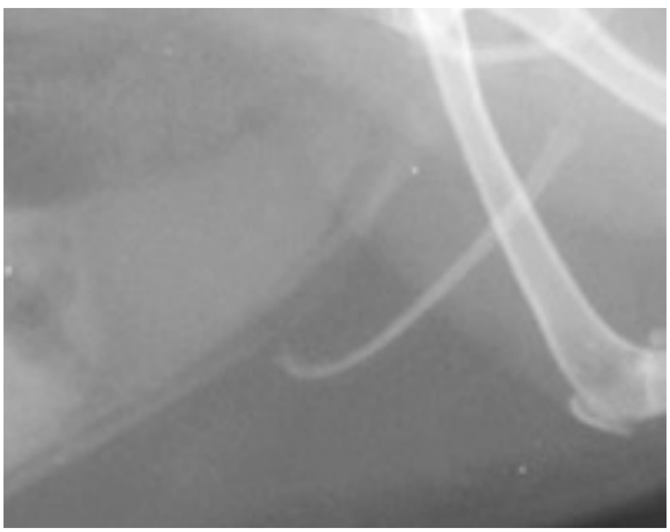
In terms of **accessory sex glands** in the male, there are a number of them to remember, please see the following illustration in the rat.



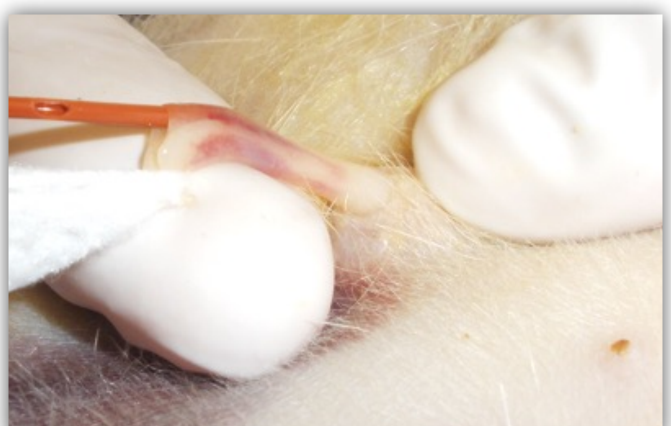
Rodents' **vesicular glands** (also called seminal glands) can be enormous, and, in some circumstances, could be confused with other anatomical structures such as a uterus. See for yourself in the picture below:



The ferret male anatomy is similar to other carnivores. There is a large **J-shaped os penis** in them.



This makes male ferrets hard to catheterize as the urethral opening is proximal to the "J hook".

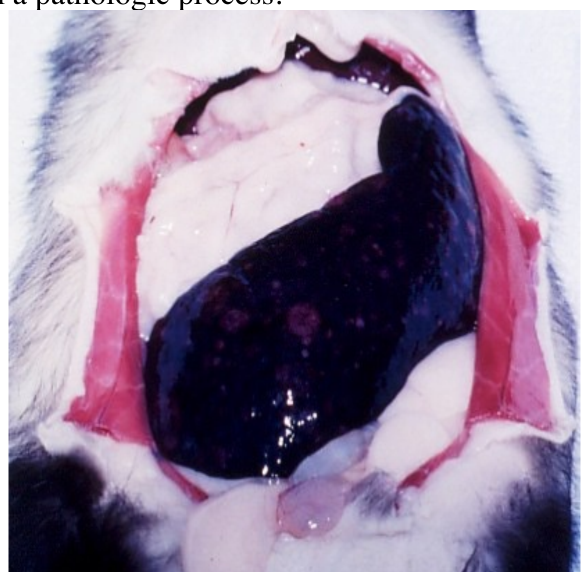


The **prostate is the main accessory gland in ferrets and completely surrounds the urethra. It can get enlarged with adrenal gland disease,** which may results in urinary obstruction (common cause of urinary obstruction in males in addition to urolithiasis). Prostatic cysts are also common in this species.

SPLEEN

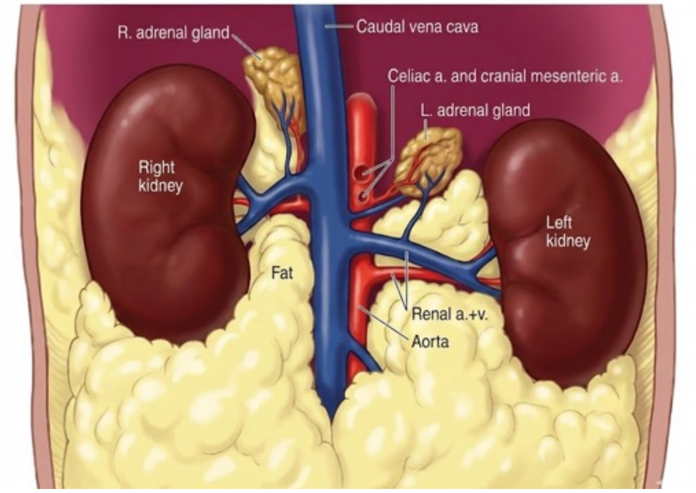
The spleen is tiny in rabbits and can be very hard to see on an abdominal ultrasound examination (also because of the gas present in the stomach).

On the flip side, it is typically giant in ferrets. Extramedullary hematopoiesis is common in this species and participates in making the spleen so large in ferrets. A a large spleen in ferrets is not, in most cases, resulting from a pathologic process.

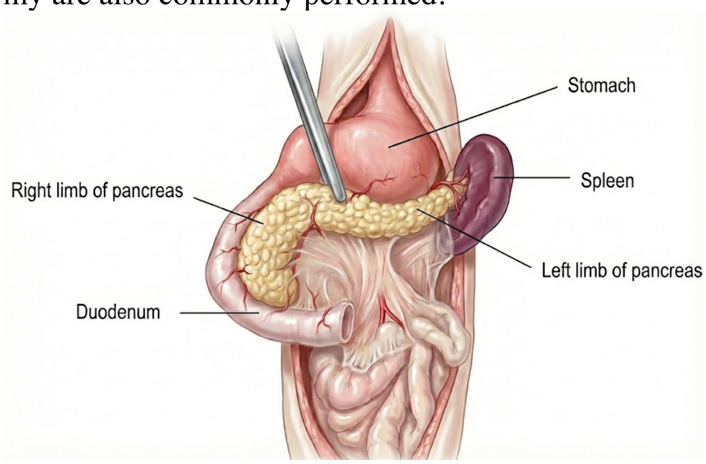


ENDOCRINE SYSTEM

There are a number of possible endocrine issues in companion small mammals. However, none is more common than **adrenal gland disease in ferrets, which is caused by hyperplasia or adrenal tumors**. It typically causes bilateral hair loss along the body. While surgery is not necessarily the first line of treatment, it is important to be aware of the anatomy of the adrenal gland as it is relevant for surgery and imaging. The **right adrenal gland is more cranial than the left. The big thing is that the right gland is very close to the caudal vena cava.** This essentially results in 2 things: 1) it will be hard to remove during surgery, harder than the left and 2) it may invade the caudal vena cava when tumorizing which may impede blood flow and make it even harder to remove (see surgery lecture).



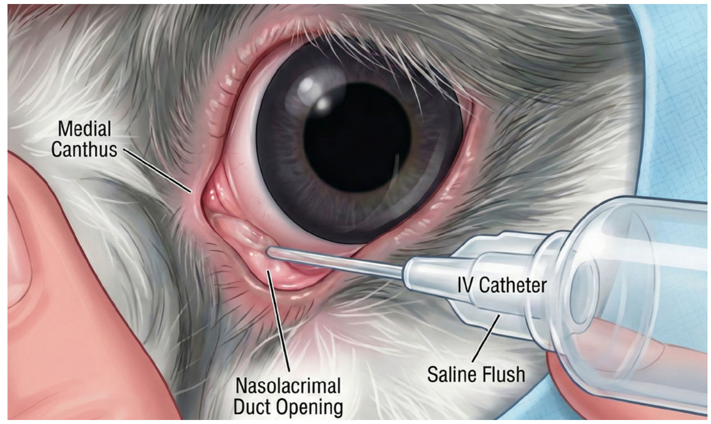
Another common endocrine disease of ferrets is **insulinoma**. It is a common cause of seizures and lethargy. The ferret pancreas is V-shaped and is located along the duodenum. It has **2 limbs, the right (which is longer) and the left**, which are connecting near the gastric pylorus. Insulinomas can be diffuse or nodular in the pancreas. Initial treatment is mainly medical, but surgical treatment such as nodulectomy or partial pancreatectomy are also commonly performed.



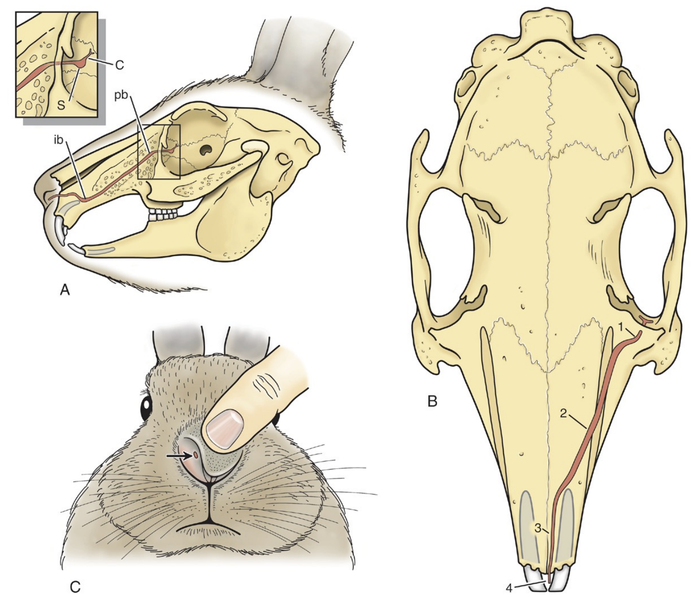
EYES

The eyes are very laterally placed in rabbits, rodents, and even in ferrets. So rabbits and some rodents have close to a 360 degree field of vision.

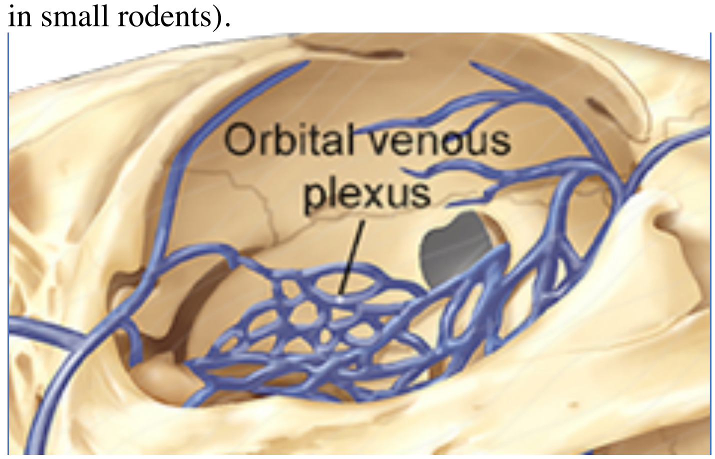
In rabbits, there is a single ventral lacrimal punctum (like in pigs, but unlike most other mammals which have two. Random elephant fact here: no lacrimal puncta in elephant species). It can be catheterize to flush it out if obstructed (and you will do exactly that during the lab).

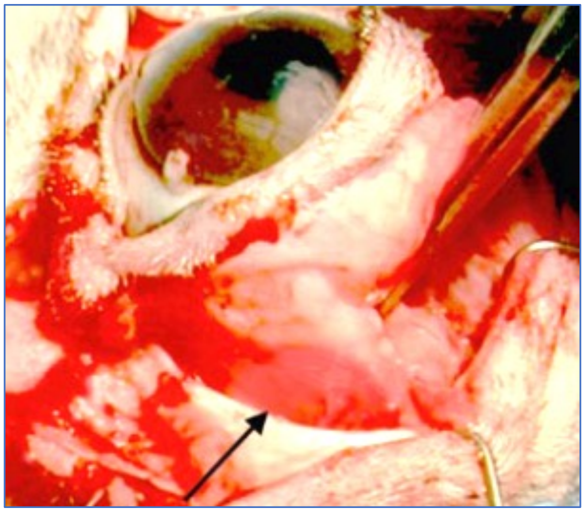


The lacrimal duct of rabbits is also in close association ventrally with the apex of the incisors (there is a bent there too) and the first maxillary premolars (P2). This means that it can get occluded with retrograde reserve crown elongation of these teeths.



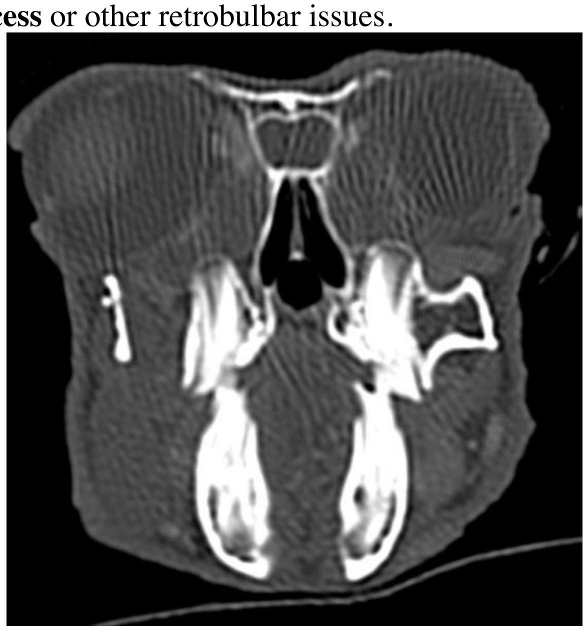
There is also a **large retrobulbar venous sinus in rabbits and rodents.** This is important because enucleation is harder to do in rabbits and rodents as you need to be extremely careful not to rupture this sinus, which can lead to a **severe hemorrhage** (or even deaths in small rodents).



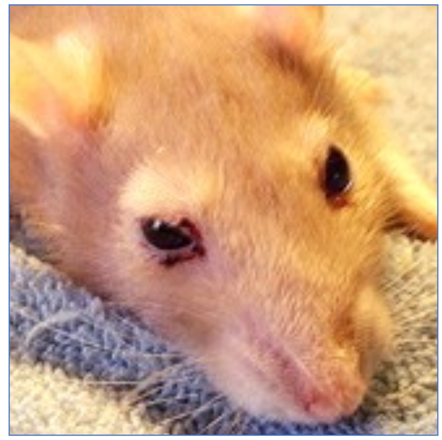


This sinus may engorge significantly with impaired blood flow to the cranial vena cava such as when a thymoma is present (which will results in exophthalmia, a common sign of thymomas) or also with heart disease.

In rabbits (and also in rodents), the maxillary cheek teeth (the 3 molars on either side) apex is located close to the retroorbital space. **With retrograde elongation of the reserve crowns, this can lead to retrobulbar abscess** or other retrobulbar issues.

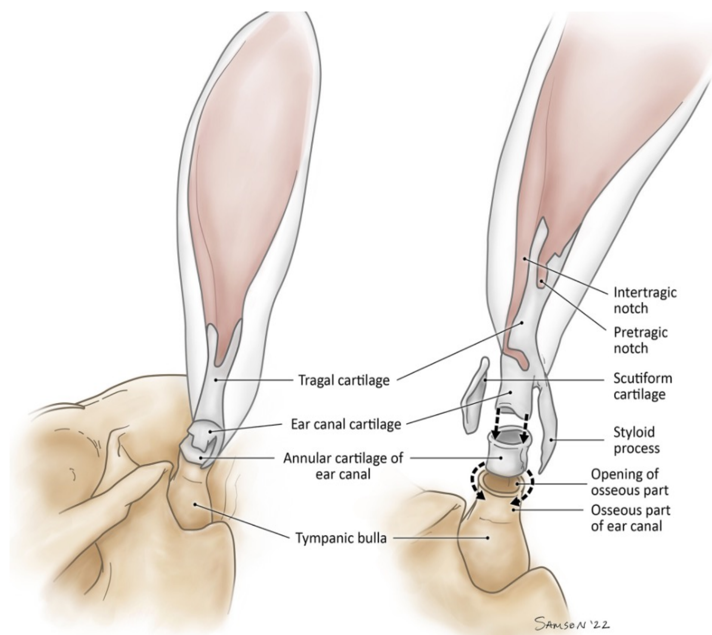


Rodents, unlike rabbits, have rudimentary third eyelids. The eyes are kind of mobile in the orbit and can frequently **appear exophthalmic and are prone to being proptosed** (it is actually one way of enucleating them, inducing proptosis surgically prior to removal, don't tell the opthhalmologists I presented that as an option). Another interesting snippet of information in rodents is that they can excrete porphyrins in their tears, which is known as **chromodacryorrhea**. With a lack of grooming, porphyrins can accumulate periocularly and look like dry blood. Viral infection by the **sialodacryoadenitis virus** (a darn coronavirus) will increase chromodacryorrhea.



EARS

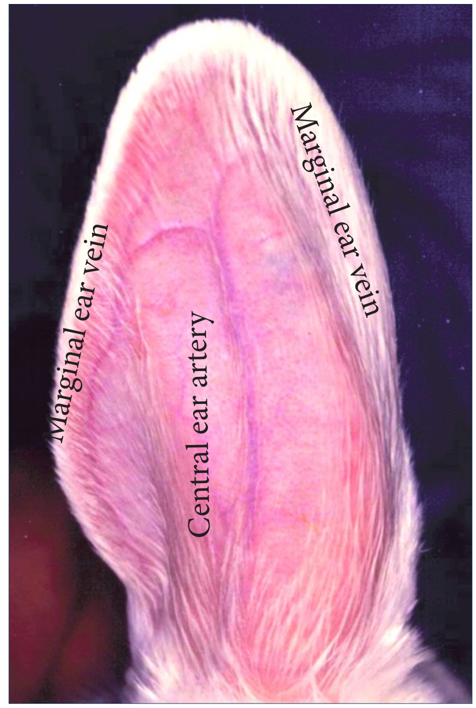
We'll mainly talk about rabbits in this section for obvious reasons. So, if you have not noticed before, rabbits have kind of big ears that are vertically erect (except in lops). There is **no separate vertical and horizontal external ear canal in rabbits**, just a single canal. There are 3 ear cartilages, the auricular cartilage (that includes the tragus and helix), the small scutiform cartilage, and the annular cartilage. When you do an otoscopic examination (during the lab), you will notice that there is a small **diverticulum laterally supported by the tragus**, you will need to redirect the otoscope to go to the external ear canal.



In lop-eared rabbits, the external canal is stenotic compressed at the point of bending because of a gap between the annular and tragal part of the au-

ricular/tragal cartilages, leading to further stenosis and predisposing these poor products of our craving for cuteness to horrible otitis.

In rabbits with big ears (so not the dwarf breeds), there are a number of prominent vessels that can be used for blood sampling and catheterization. The **central ear artery** can be used for palpating pulse and placing intraarterial catheters and the **marginal ear veins** can be used for venipuncture and placing intravenous catheters. There is **extensive arteriovenous anastomosis** in the rabbit's ears and, partly for this reason, **the ears are prone to thrombosis and necrosis** when injecting some drugs or even with just catheterization. It is always a risk that should be discussed with the owner should you elect to place a catheter at that location.



The Ten Most Important Facts

1. Rabbits have thin skin and should be shaved carefully prior to surgery or other procedures. Also don't shave the plantar surface of the pelvic limbs.
2. Rabbits have a light skeleton, well developed epaxial and hindlimb muscles that predispose them to vertebral fractures and spinal trauma.
3. Rabbits and rodents are monogastric hindgut fermenters. They have a well-developed cecum where bacterial fermentation occurs resulting in the production of volatile fatty acids. Rabbits have a "wash-back"mechanism in their colon to selectively retain small food particle and fluid in their cecum while rodents have a "mucus trap" to recycle bacteria.
4. Rabbits and rodents produce two types of feces with one type being nitrogen-rich and called cecotrophes. Cecotrophy is particularly effective in rabbits.
5. Rabbits and rodents are obligate nasal breathers. Any lesion or disease of the upper airways will lead to dyspnea. They are also harder than other domestic mammals to intubate because of their pharyngeal anatomy, especially in rodents that have a palatal ostium.
6. The rabbit's thymus persists into adulthood. Rabbits are prone to thymoma, which lead to respiratory disease and cranial caval syndrome.
7. Rabbits have a markedly different calcium metabolism in that intestinal absorption is unregulated and directly affects plasma calcium and the urinary fractional excretion of calcium is very high, sometimes resulting in bladder "sludge".
8. Rats and mice have extensive mammary tissue and, as a result, mammary tumors may manifest anywhere along the trunk and proximal limbs.
9. Rabbits have a single ophthalmic punctum on the lower eyelid and their nasolacrimal duct is in close proximity to the maxillary incisors and first upper premolars (P2).
10. Rabbit's external ear canal is a single tube with no separation into horizontal and vertical portions. Lop-eared rabbits have a stenotic ear canal and are therefore more prone to external otitis.